

Assignment Questions : 20 Marks

1. Network Application Architecture

- **Question:** Explain the concept of network application architecture. How does the client-server model differ from the peer-to-peer model in terms of communication and scalability?

2. Process Communication in the Internet

- **Question:** Describe how processes communicate over the Internet. What are the key differences between message passing and socket-based communication in computer networks?

3. Hypertext Transfer Protocol (HTTP)

- **Question:** Explain the working of HTTP with the help of an example. Discuss the difference between HTTP/1.1 and HTTP/2, focusing on performance improvements and multiplexing.

4. File Transfer Protocol (FTP)

- **Question:** Discuss the File Transfer Protocol (FTP) and its modes of operation (active vs. passive). What are the advantages and disadvantages of using FTP over other file transfer methods like HTTP or SFTP?

5. Electronic Mail in the Internet

- **Question:** Describe how email works in the Internet, detailing the role of SMTP, POP3, and IMAP in the email delivery process. How does the process differ for a web-based email service like Gmail compared to a traditional email client?

6. Short Message Peer-to-Peer Protocol (SMPP)

- **Question:** What is the Short Message Peer-to-Peer Protocol (SMPP), and how does it facilitate the exchange of SMS messages between systems? Discuss its role in modern messaging systems.

7. Transport Layer Principles: UDP vs. TCP

- **Question:** Compare and contrast UDP and TCP in terms of connection-oriented vs. connectionless communication, reliability, and performance. Provide examples of applications that would typically use each protocol.

8. Congestion Control in TCP

- **Question:** Explain the principles of TCP congestion control. How do the algorithms such as slow start, congestion avoidance, and fast recovery work together to control network congestion?

9. Network Layer and Routing Algorithms

- **Question:** Describe the differences between virtual circuit and datagram networks. Explain how routers handle packet forwarding in each model. How do different routing algorithms (e.g., Dijkstra, Bellman-Ford) play a role in routing decisions?

10. Multicast Routing in the Internet

- **Question:** What is multicast routing, and why is it important in modern network communication? Explain the concept of multicast routing protocols (e.g., DVMRP, PIM) and their use in efficient data distribution.